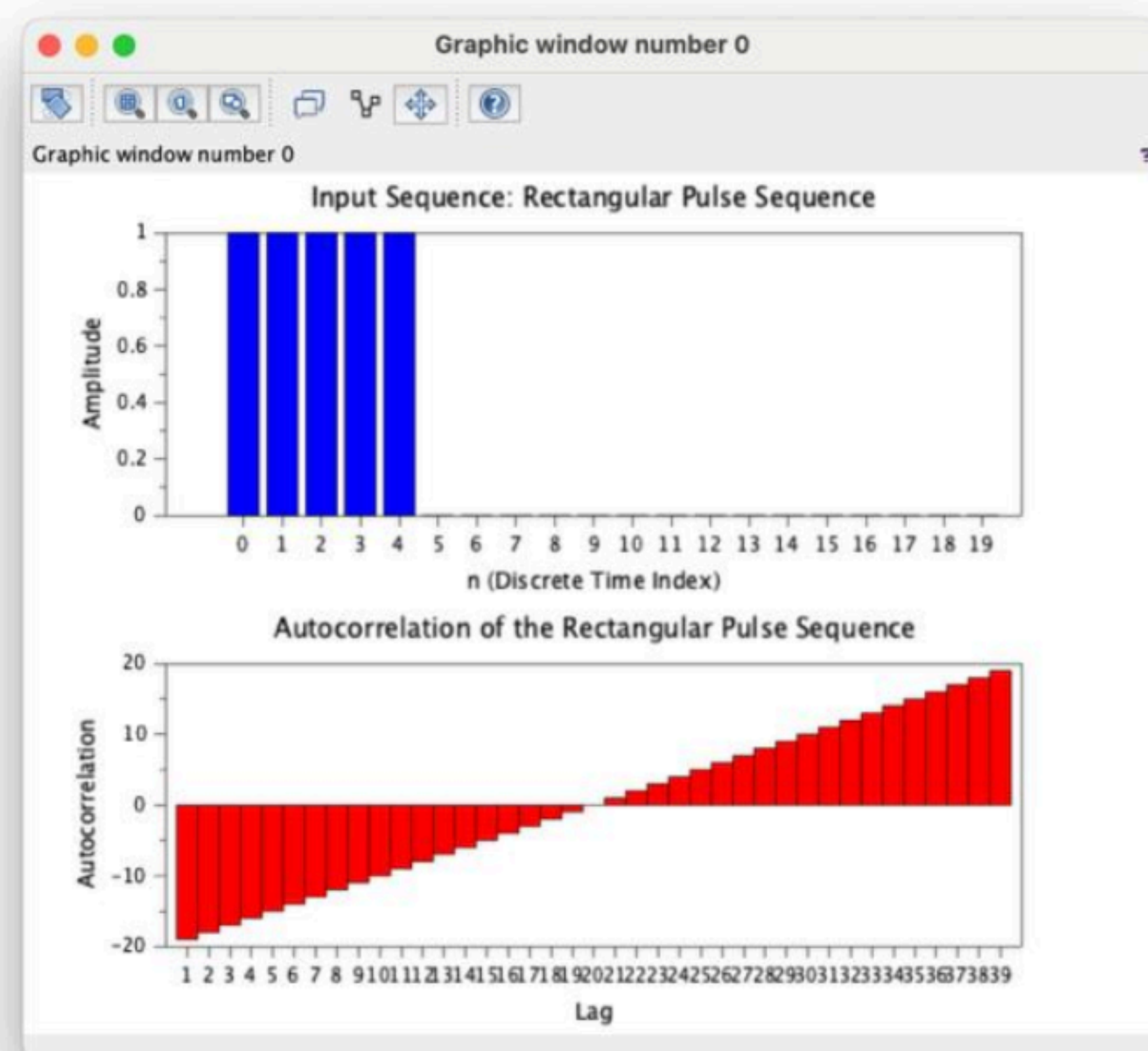


```

1 // Test Case: Autocorrelation of a Rectangular Pulse Sequence
2 // Define parameters
3 N = 20; // Number of samples
4 x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
5 // Compute the autocorrelation
6 R_xx = correl(x_n, x_n);
7 // Create a lag vector for plotting
8 lags = -N+1:N-1;
9 // Plot the input sequence
10 clf; // Clear the current figure
11 subplot(2,1,1);
12 bar(0:N-1, x_n, 'b');
13 xlabel('n (Discrete Time Index)');
14 ylabel('Amplitude');
15 title('Input Sequence: Rectangular Pulse Sequence');
16 // Plot the autocorrelation sequence
17 subplot(2,1,2);
18 bar(lags, R_xx, 'r');
19 xlabel('Lag');
20 ylabel('Autocorrelation');
21 title('Autocorrelation of the Rectangular Pulse Sequence');
22

```



```

1 // Test Case: Cross-Correlation of Two Discrete-Time Sequences
2 // Define parameters
3 N = 20; // Number of samples
4 x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
5 delay = 3; // Delay of 3 samples
6 y_n = [zeros(1,delay), x_n(1:$-delay)]; // Delayed version of x[n]
7 // Compute the cross-correlation
8 R_xy = correl(x_n, y_n);
9 // Create a lag vector for plotting
10 lags = -N+1:N-1;
11 // Plot the first sequence
12 clf; // Clear the current figure
13 subplot(3,1,1);
14 bar(0:N-1, x_n, 'b');
15 xlabel('n (Discrete Time Index)');
16 ylabel('Amplitude');
17 title('Sequence 1: Rectangular Pulse Sequence');
18 // Plot the second sequence
19 subplot(3,1,2);
20 bar(0:N-1, y_n, 'g');
21 xlabel('n (Discrete Time Index)');
22 ylabel('Amplitude');
23 title('Sequence 2: Delayed Rectangular Pulse Sequence');
24 // Plot the cross-correlation sequence
25 subplot(3,1,3);
26 bar(lags, R_xy, 'r');
27 xlabel('Lag');
28 ylabel('Cross-Correlation');
29 title('Cross-Correlation between Sequence 1 and Sequence 2');
30

```

